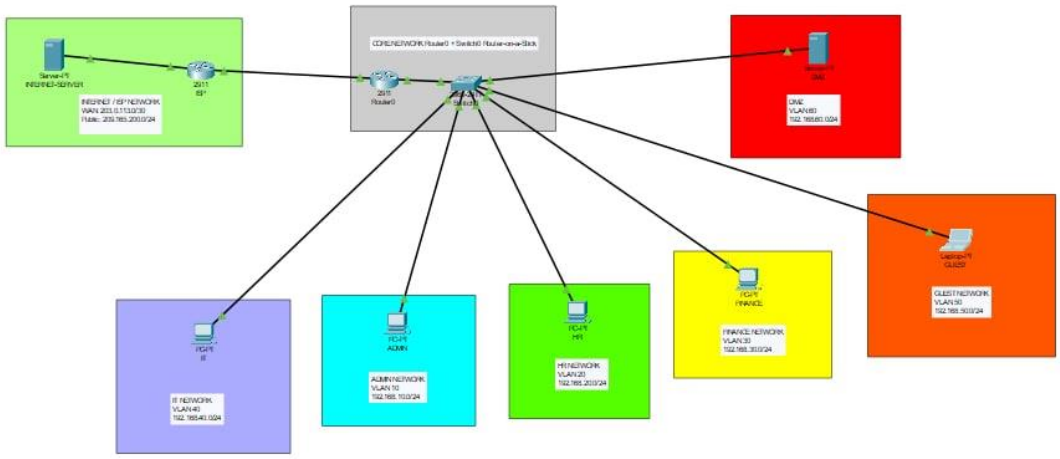
PWDbyMe SecureNet Lab

Secure Enterprise Network Simulation



Cisco Packet Tracer topology used for the lab.

Documentation

Cisco Packet Tracer | VLAN Segmentation | ACLs | SSH | Port Security | NAT/PAT | DMZ

This document records the network design, major configuration commands, validation tests, and observations from the hands-on lab. Lab credentials are intentionally omitted or redacted.

1. Project Overview

Objective. Build and secure a simulated enterprise network by separating business functions into VLANs, controlling inter-network access, hardening switch ports, restricting device management, providing Internet connectivity through NAT/PAT, and isolating a public-facing web server in a DMZ.

Core devices. Cisco 2911 enterprise router, Cisco 2960 switch, endpoint PCs for Admin/HR/Finance/IT, Guest laptop, DMZ server, ISP router, and an Internet-side server.

Implemented Security Controls

Control	Purpose
VLAN segmentation	Separates departments and security zones at Layer 2.
Router-on-a-Stick	Provides controlled inter-VLAN routing through router subinterfaces.
Extended ACLs	Blocks selected cross-VLAN traffic such as Guest-to-internal and HR-to-Finance.
SSH management	Replaces insecure Telnet-style management and limits remote administration to IT.
Switch port security	Restricts access ports to one learned MAC address using sticky MAC learning.
Unused-port shutdown	Reduces exposure from unused physical switch ports.
NAT/PAT	Translate private internal addresses to the R1 WAN address for simulated Internet access.
DMZ + Static PAT	Publishes the DMZ HTTP service without exposing internal user VLANs directly.

VLAN and Addressing Plan

VLAN	Zone	Subnet	Gateway / Key Host
10	ADMIN	192.168.10.0/24	192.168.10.1
20	HR	192.168.20.0/24	192.168.20.1
30	FINANCE	192.168.30.0/24	192.168.30.1
40	IT	192.168.40.0/24	192.168.40.1
50	GUEST	192.168.50.0/24	192.168.50.1
60	DMZ	192.168.60.0/24	192.168.60.1 / Server 192.168.60.10
99	MANAGEMENT	192.168.99.0/24	R1 192.168.99.1 / SW1 192.168.99.2
WAN	R1 - ISP	203.0.113.0/30	ISP .1 / R1 .2
Public	Internet segment	209.165.200.0/24	Internet Server 209.165.200.10

2. VLAN Segmentation and Switch Configuration

The access switch was divided into separate VLANs for each department and security zone. Access ports Fa0/2 through Fa0/7 were assigned to the corresponding VLANs, while Fa0/1 was configured as the trunk toward R1.

```
enable configure
terminal vlan 10
name ADMIN vlan
20  name HR vlan
30  name FINANCE
vlan 40  name IT
vlan 50  name
GUEST vlan 60
name DMZ vlan 99
name MANAGEMENT

interface fa0/2
switchport mode access
switchport access vlan 10
interface fa0/3
switchport mode access
switchport access vlan 20
interface fa0/4
switchport mode access
switchport access vlan 30
interface fa0/5
switchport mode access
switchport access vlan 40
interface fa0/6
switchport mode access
switchport access vlan 50
interface fa0/7
switchport mode access
switchport access vlan 60
interface fa0/1
switchport mode trunk
```

```
VLAN Name                              Status    Ports
---- --------------------------------- --------- -------------------------------
1    default                           active    Fa0/1, Fa0/8, Fa0/9, Fa0/10
                                                 Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                 Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                 Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10   ADMIN                             active    Fa0/2
20   HR                                active    Fa0/3
30   FINANCE                           active    Fa0/4
40   IT                                active    Fa0/5
50   GUEST                             active    Fa0/6
60   DMZ                               active    Fa0/7
99   MANAGEMENT                        active
1002 fddi-default                      active
1003 token-ring-default                active
1004 fddinet-default                   active
1005 trnet-default                     active
Switch#
```

Validation: show vlan brief confirmed Fa0/2-Fa0/7 were assigned to VLANs 10-60.

Observation: The VLAN table confirmed that each user endpoint was placed in the intended broadcast domain. Fa0/1 did not appear as an access member because it was operating as a trunk.

Router-on-a-Stick

R1 sub interfaces were created on G0/0 so each VLAN had a Layer 3 gateway. The parent interface remained unnumbered, and the sub interfaces used 802.1Q encapsulation.

```
interface g0/0
no shutdown

interface g0/0.10

 encapsulation dot1Q 10  ip address
192.168.10.1 255.255.255.0 interface
g0/0.20  encapsulation dot1Q 20  ip
address 192.168.20.1 255.255.255.0
interface g0/0.30  encapsulation
dot1Q 30  ip address 192.168.30.1
255.255.255.0 interface g0/0.40
encapsulation dot1Q 40  ip address
192.168.40.1 255.255.255.0 interface
g0/0.50  encapsulation dot1Q 50  ip
address 192.168.50.1 255.255.255.0
interface g0/0.60  encapsulation
dot1Q 60  ip address 192.168.60.1
255.255.255.0 interface g0/0.99
encapsulation dot1Q 99
 ip address 192.168.99.1 255.255.255.0
```

Observation: All configured sub interfaces reported up/up. Pings from Admin to its gateway succeeded, and cross-VLAN pings succeeded before restrictive ACLs were applied. A first ICMP packet occasionally timed out while ARP resolution completed, then subsequent replies succeeded.

3. Access Control Lists (ACLs)

Extended ACLs were placed inbound on the source VLAN sub interfaces so unwanted traffic could be denied close to its origin.

Guest Network Policy

```
ip access-list extended GUEST-IN  deny ip
192.168.50.0 0.0.0.255 192.168.10.0 0.0.0.255  deny
ip 192.168.50.0 0.0.0.255 192.168.20.0 0.0.0.255
deny ip 192.168.50.0 0.0.0.255 192.168.30.0
0.0.0.255  deny ip 192.168.50.0 0.0.0.255
192.168.40.0 0.0.0.255  deny ip 192.168.50.0
0.0.0.255 192.168.99.0 0.0.0.255  permit ip
192.168.50.0 0.0.0.255 any

interface g0/0.50  ip
access-group GUEST-IN in
```

HR Policy

```
ip access-list extended HR-IN  deny ip 192.168.20.0
0.0.0.255 192.168.30.0 0.0.0.255  permit ip
192.168.20.0 0.0.0.255 any

interface g0/0.20  ip
access-group HR-IN in
```

Source	Destination	Result
Guest VLAN 50	Admin / HR / Finance / IT / Management	BLOCKED
Guest VLAN 50	DMZ / Internet	ALLOWED
HR VLAN 20	Finance VLAN 30	BLOCKED
HR VLAN 20	Admin / DMZ	ALLOWED
IT VLAN 40	Internal networks	ALLOWED

Observation: The ACLs changed the network from unrestricted inter-VLAN routing to policy-controlled routing. Guest hosts could still reach approved external/DMZ services while protected internal VLANs were unreachable.

4. Secure Device Management with SSH

SSH was configured on both R1 and SW1. Local credentials were created with privilege 15, RSA keys were generated, SSH version 2 was enabled, and VTY lines were restricted to SSH. The management VLAN provided dedicated management addresses.

```
hostname R1 ip domain-name pwdbyme.local
username netadmin privilege 15 secret
<REDACTED> enable secret <REDACTED> crypto key
generate rsa ip ssh version 2

line vty 0 4  login
local  transport
input ssh

access-list 10 permit 192.168.40.0
0.0.0.255 line vty 0 4  access-class 10 in

! SW1 management SVI interface vlan
99  ip address 192.168.99.2
255.255.255.0  no shutdown ip
default-gateway 192.168.99.1

! SW1 VTY restriction access-list 10 permit
192.168.40.0 0.0.0.255 line vty 0 15

 login local
transport input ssh
access-class 10 in
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ssh -l netadmin 192.168.99.1

Password:



R1#exit

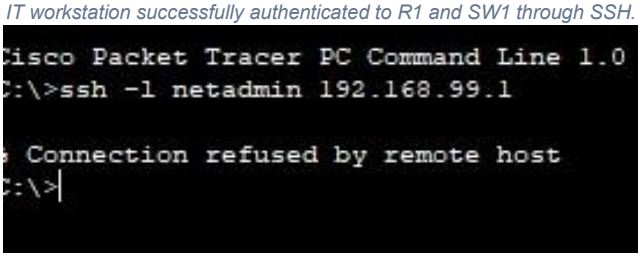
[Connection to 192.168.99.1 closed by foreign host]
C:\>ssh -l netadmin 192.168.99.2

Password:



SW1#en
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#e
% Ambiguous command: "e"
SW1(config)#ex
SW1#end
Translating "end"...domain server (255.255.255.255)
```

IT workstation successfully authenticated to R1 and SW1 through SSH.



Admin workstation test: SSH connection was refused, confirming the IT-only VTY access restriction.

Observation: Remote management worked from the IT VLAN but was refused from Admin. This demonstrated management-plane access control independent of ordinary routed reachability.

5. Switch Port Security and Physical-Layer Hardening

Port security was enabled on access ports Fa0/2-Fa0/7. Each port was limited to one sticky-learned MAC address, and the violation action was set to shutdown.

```
interface fa0/2  switchport mode access
switchport port-security  switchport port-
security maximum 1  switchport port-
security mac-address sticky  switchport
port-security violation shutdown

! Same policy applied to Fa0/3 through
Fa0/7
```

The violation test used the Guest access port. After the authorized Guest laptop generated traffic and its MAC address was learned, another endpoint was connected to the same switch port and attempted to ping the gateway.

Observation: The unauthorized endpoint triggered a port-security violation and the port entered secure-shutdown. The port was recovered by reconnecting the authorized device and issuing shutdown followed by no shutdown.

Unused Port Shutdown

```
interface range fa0/8 -
24  shutdown  description
UNUSED_PORT

interface range gig0/1 - 2
shutdown
 description UNUSED_PORT
```

Unused access interfaces were administratively disabled to reduce the opportunity for unauthorized physical connections.

6. ISP Simulation and NAT/PAT

R1 was connected to a simulated ISP over 203.0.113.0/30. The Internet-side server used 209.165.200.10/24. R1 used a default route toward the ISP and PAT so multiple private hosts could share the R1 WAN address 203.0.113.2.

```
interface g0/1  ip address 203.0.113.2
255.255.255.252  no shutdown  ip nat
outside

ip route 0.0.0.0 0.0.0.0 203.0.113.1

interface g0/0.10
ip nat inside
interface g0/0.20
ip nat inside
interface g0/0.30
ip nat inside
interface g0/0.40
ip nat inside
interface g0/0.50
ip nat inside
interface g0/0.60
ip nat inside
interface g0/0.99
ip nat inside

access-list 1 permit 192.168.0.0 0.0.255.255 ip
nat inside source list 1 interface g0/1 overload
```

```
R1#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.2:1032  192.168.50.10:9    209.165.200.10:9   209.165.200.10:1032
icmp 203.0.113.2:1033  192.168.50.10:10   209.165.200.10:10  209.165.200.10:1033
icmp 203.0.113.2:1034  192.168.50.10:11   209.165.200.10:11  209.165.200.10:1034
icmp 203.0.113.2:1035  192.168.50.10:12   209.165.200.10:12  209.165.200.10:1035
icmp 203.0.113.2:13    192.168.50.10:13   209.165.200.10:13  209.165.200.10:13
icmp 203.0.113.2:14    192.168.50.10:14   209.165.200.10:14  209.165.200.10:14
icmp 203.0.113.2:15    192.168.50.10:15   209.165.200.10:15  209.165.200.10:15
icmp 203.0.113.2:16    192.168.50.10:16   209.165.200.10:16  209.165.200.10:16
```

show ip nat translations displayed 192.168.50.10 translated through 203.0.113.2 while reaching 209.165.200.10.

Observation: Guest Internet traffic succeeded even though Guest-to-internal access remained blocked. The translation table confirmed that PAT was operational and the internal address was being represented externally by the R1 WAN address.

7. DMZ Web Server and Public HTTP Access

A web server was placed in VLAN 60 with private address 192.168.60.10. HTTP was enabled on the server and the page was customized as a PWDbyMe SecureNet status dashboard. Static PAT mapped the R1 public HTTP port to the DMZ server. `ip nat inside source static tcp 192.168.60.10 80 203.0.113.2 80`

```
! Optional HTTPS mapping if enabled
ip nat inside source static tcp 192.168.60.10 443 203.0.113.2 443
```

The public-side test browsed to **http://203.0.113.2**. The request was translated to 192.168.60.10:80 inside the DMZ.

Observation: The public web service could be reached through the WAN address without assigning a public address directly to the DMZ server. Internal user VLANs were not published as public services.

8. Final Validation Matrix

Test	Expected	Observed Result
Admin -> 192.168.10.1	Allowed	Ping successful
Admin -> HR before ACL restriction	Allowed	Inter-VLAN routing successful
Guest -> Admin/HR/Finance/IT	Blocked	ACL denied access
Guest -> Internet Server	Allowed	Ping successful through NAT/PAT
HR -> Finance	Blocked	ACL denied access
IT -> R1 via SSH	Allowed	SSH login successful
IT -> SW1 via SSH	Allowed	SSH login successful
Admin -> R1 via SSH	Blocked	Connection refused by remote host
Unauthorized endpoint -> secured port	Blocked	Port security violation / secure-shutdown
Internet side -> 203.0.113.2:80	Allowed	DMZ web page reachable via static PAT

9. Key Observations and Learning Outcomes

1. **Segmentation alone does not enforce business policy.** VLANs separated broadcast domains, but ACLs were required to decide which routed communications were permitted.

2. **Management-plane access can be restricted separately.** IT-only VTY ACLs allowed the IT workstation to manage network devices while Admin SSH attempts were rejected.

3. **Port security protects the access layer.** Sticky MAC learning and shutdown-on-violation provided a simple control against unauthorized endpoint replacement on a secured switch port.

4. **NAT/PAT provides address translation, not segmentation.** Guest hosts could access the simulated Internet through PAT while internal ACL policy continued to block protected VLANs.

5. **A DMZ limits direct exposure.** The public HTTP service was published through static PAT while the server remained on a separate private VLAN.

Result: The final lab combined segmentation, routing, policy enforcement, secure administration, access-layer hardening, address translation, and DMZ publishing into one enterprise-style Packet Tracer topology.

10. Useful Verification Commands

```
show vlan brief show interfaces
trunk show ip interface brief
show access-lists show port-
security show port-security
interface fa0/6 show port-
security address show interfaces
status show ip nat translations
show ip nat statistics show
running-config
```

These commands were used throughout the lab to verify VLAN membership, interface state, ACL matches, port-security status, and NAT/PAT behavior.